

OpenNFCSense: Open-Source Library for NFCSense

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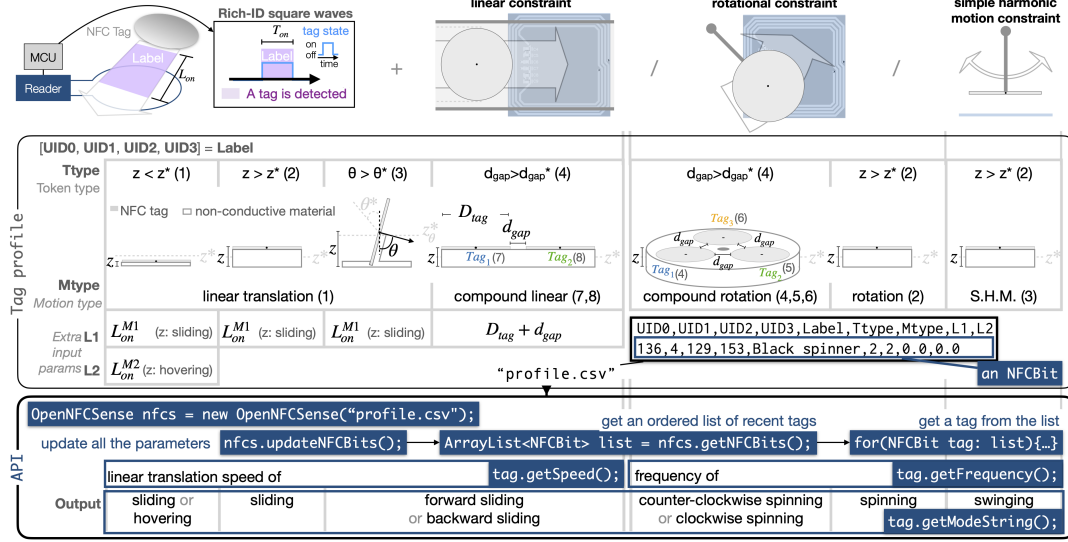


Figure 1: OpenNFCSense is an open-source library for NFCSense [2]. With a user-defined tag profile, the users can use its API to obtain the tagged objects' motion features that a commodity NFC reader measures.

ABSTRACT

OpenNFCSense is an open-source library for NFCSense, a system for detecting the movements of near-field communication (NFC) tags using a commodity, low-cost RC522 NFC reader. With a user-defined tag profile, the users can use its application programming interface (API) to obtain the NFC tagged objects' motion speed, motion frequency, and motion type while recognizing these tagged objects. Since NFC tags support straightforward augmentation to existed physical objects, such as modular building blocks and mechanical toys, OpenNFCSense offers an easy and safe method for rapid prototyping tangible user interfaces for designers.

CCS CONCEPTS

• Human-centered computing → User interface toolkits; • Hardware → Sensor applications and deployments.

KEYWORDS

NFCSense, API, rapid prototyping, tangible interaction

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1 INTRODUCTION

NFCSense [2] is a system of data-defined rich-ID motion-sensing techniques using a single commodity near-field communication (NFC) reader and tags. By running the reader at a high refresh rate (e.g., 300 reads per second) and using it with a physical constraint, the system can measure the motion speed, frequency, and types of moving NFC tag(s). Furthermore, it supports various hands-off, multi-tag inputs and can be further used as a wearable user interface. The feasibility of using an NFC reader as a motion sensor has been validated, so the next objective is to support designers and developers to implement tangible user interfaces [1] with NFCSense.

This work introduces OpenNFCSense, an open-source library for NFCSense. OpenNFCSense provides high-level application programming interface (API), as shown in Figure 1. The latest version of this library can be retrieved from the Github repository¹.

Prerequisites and Installation. OpenNFCSense library requires simple, readily available hardware such as an Arduino Uno board, a

¹<https://github.com/howieliang/OpenNFCSense>

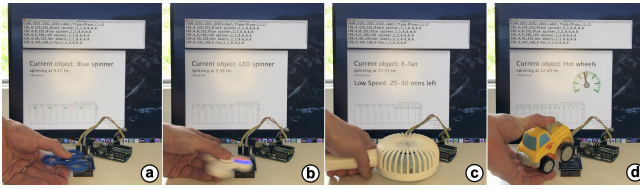


Figure 2: Hot-swappable sensing of the rotation frequency of tagged objects with a single tag profile.

low-cost RC522 NFC reader, and a few NFC tags, as well as a PC or laptop with Arduino² and Processing³ development environment installed. A user can install the OpenNFCSense library to the Processing software, connect the RC522 NFC reader to the Arduino board, and upload the Arduino codes to the Arduino board when the reader is connected. Then, the user can import the library and use the APIs with the hardware.

Editing The Tag Profile. OpenNFCSense provides several readily available example sketches that demonstrate the uses of its API, including a boilerplate for application development. In the data folder of each example sketch, there is a comma-separated value file of tag profile (e.g., profile.csv) with the following headers in each column:

UID0,UID1,UID2,UID3,Label,Ttype,Mtype,L1,L2

where

- *UID*[0 – 3] is the 4-byte UID of the tag.
- *Label* is the user-defined name of the tag.
- *Ttype* is the token type.
- *Mtype* is the motion type.
- *L1*, *L2* are extra parameters (unit: mm).

Users can bind a new NFC tag to an example code by adding a new row to the file as the tag profile of an *NFCBit*. The user first runs the example, scans the tag with the reader, and sees the tag's 4-byte UID from the console. Then, the user duplicates an existed row and replaces the old UID and name with the new ones. If necessary, The user also modifies other column values in the row according to the token type they made and the motion type they want to detect, as depicted in Figure 1. After the user saved the profile and reruns the example, the example sketch can recognize the tag and its corresponding motion.

Sensing The Motion Frequency and Sequence of Tags. The easiest applications are sensing the motion frequency and the tag orders because they do not require extra parameters (e.g., tag activation path distance). A user can use *getFrequency()* to get the rotation or swinging frequency of a spinning object and use *getName()* to get what tag is, as the results shown in Figure 2. A user can use *getNFCBits()* to get an ordered list of recent tags so that they can utilize the tag sequences, as the results shown in Figure 3.

Sensing The Motion Speed and Types. The users can obtain the linear translation speed and types (hovering or direction) of a tagged object with extra parameters specified in the profile. A single tag

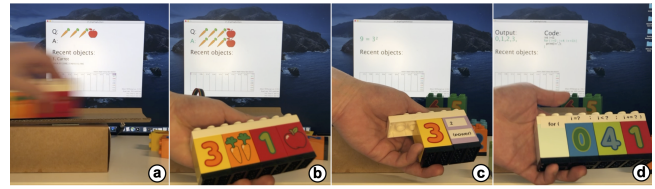


Figure 3: Sensing the sequence of tagged building blocks for education applications using (a-b) existed blocks, (d) self-defined blocks, and (c) self-defined constraints.

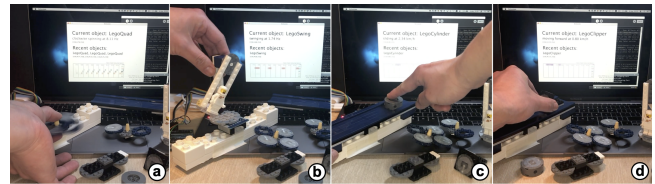


Figure 4: NFCBits and physical constraints that are made with lego bricks: (a) a spinner; (b) a swing; (c) a clipper ship mounted a tilted tag; (d) a flippable cylinder.

requires the tag activation path distance obtained by manual measurement with a simple apparatus, such as a ruler and a track. A compound tag requires the tag coil diameter and the gap distance between tags. A user can use *getMode()* or *getModeString()* to get the index or name of the motion type and use *getSpeed()* to get the linear translation speed accordingly.

We have been building up an activation distance database⁴ for pairs of commodity NFC tag stickers (e.g., NTAG213) and readers (e.g., RC522) so that the users are no longer required to enter tag activation path distance in the linear speed sensing. The next steps are to extend the dataset and make a more user-friendly editor that blends the dataset into the tag profile configuration.

Making NFCBits with Existed Objects and Modular Building Blocks. Commodity NFC tag stickers (e.g., NTAG213) can straightforwardly augment already-existing objects, such as figurines and mechanical toys. Therefore, the TUI designers can exploit and leverage the rich semantics and mechanics of existed everyday objects in human-computer interaction, as shown in Figure 2. Exceptions are conductive or highly dielectric objects that could affect the NFC antenna coupling and thus deteriorate or block the link of signal transmission.

Users who do not have a fabrication tool or machine at their workspaces can build their NFCBits using modular building blocks, such as Lego bricks and Duplo blocks, to achieve high precision and replicability. The users can reuse the previously obtained parameters (i.e., tag activation path distance) among the built profiles that they made once. Figure 3 and 4 show examples of NFCBits and physical constraints that are made of only worldwide available modular building blocks, which are not only easy to fabricate but also can be reproduced anywhere.

²<https://www.arduino.cc/>

³<https://processing.org/>

⁴<https://github.com/howieliang/NFCSense>

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